

5(a)	<u>embryo</u> ;	1
5(b)(i)	any two from: 1 as water potential increases the germination index increases ; ora 2 water potential in MPa and index figure given for two points ;	2
5(b)(ii)	any three from: 1 low / negative, water potential decreases water, uptake / in, seed ; ora 2 water, needed / used, to activate embryo / produce gibberellin ; 3 water, needed / used, for hydrolysis (reactions) ; 4 starch to maltose / maltose to glucose ; 5 water, needed / used, as a medium for reactions ; 6 solutes (lowering water potential) may, be toxic / inhibit enzymes ; 7 AVP ;	3
5(c)	<i>enzyme</i> amylase / maltase / protease ; <i>location</i> aleurone layer ;	2
5(d)(i)	less, energy / resources, to make stem so, more to / larger, <u>flowers</u> or less likely to be damaged by, wind / storm / heavy rain or takes less, water / fertiliser / time to flower ;	1

Question	Answer	Marks
5(d)(ii)	any three from: <i>short / dwarf, plants:</i> 1 are, homozygous recessive / <i>lele</i> ; 2 <i>le</i> / recessive allele, codes for, non-functional / no, enzyme ; 3 <i>le</i> / recessive allele / non-functional enzyme, gives inactive gibberellin ; 4 DELLA proteins, not broken down / stay bound to PIF ; 5 PIF cannot, bind to promoter / start transcription ; 6 of gene(s) that promote, growth / cell division / cell elongation ;	3

Question	Answer	Marks
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